HANNAH NAGPALL

PhD Applicant · Trustworthy AI for Healthcare & High-Stakes Decision-Making

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RESEARCH INTERESTS

I work on **change control for agentic clinical AI**: when a model or its harness (prompts, memory, tools, stopping rules) is updated, how do we know whether patient outcomes changed, why, and what should change next? My 4 papers (2 accepted at AI4S 2026, 2 under review at SN Computer Science) include 1 first-author, 2 sole-author and 1 co-authored paper, plus a first-author abstract accepted at AIChE 2026; code is released openly. My work covers **memory-induced bias in LLM health agents** (45.1% to 60.0% accuracy on MIMIC-IV, with a null result on a second model), **unstable explanations** (BiLSTM explanations diverging across seeds at near-identical accuracy), and **privacy auditing**. I report negative results as such and measure what a system does without an update before trusting it, a habit built in Raman-spectroscopy fuel analysis. I seek a PhD developing methods to detect, attribute and clinically review update-driven changes on longitudinal patient data.

EDUCATION

Texas A&M University–Kingsville, USA — B.S. Computer Science, Honors College — Aug 2025 – May 2027 (exp.)

- GPA **3.93 / 4.0**, top 5% of class · Honors College Scholar · Presidential List (Spring 2026) · Dean's List (Fall 2025)

Alabama A&M University, USA — B.S. Computer Science (transferred) — Aug 2023 – May 2025

- GPA **3.95 / 4.0**, top 2% of class · Presidential Silver Medallion (Spring 2025) · Presidential Freshman Award

PUBLICATIONS

Accepted

Memory That Remembers Wrong: Quantifying and Mitigating Confirmation-Loop Bias in Self-Evolving LLM Health Agents

Hannah Nagpall (first author), et al. · AI4S Conference 2026 · Code

- Tested whether accumulated memory entrenches early diagnostic errors, using 50 real MIMIC-IV v3.1 patient trajectories (872 predictions) with objective KDIGO-derived ground truth. Counter-evidence prompting raised final-encounter accuracy from **45.1% to 60.0%** (exact McNemar $p = 0.021$; 9 of 10 discordant pairs; bootstrap 95% CI on the gain 0.04–0.28).
- A three-arm ablation showed unmitigated memory was no better than no memory ($p = 0.80$), isolating the gain to the mitigation. The effect did **not** replicate on Qwen3.6-27B ($p = 1.0$); reported as a backbone-dependent null result.

The Faithfulness Paradox: When a Simple Classifier Out-Explains Neural Networks in Hate Speech Detection

Hannah Nagpall (co-author), et al. · AI4S Conference 2026

- Compared TF-IDF logistic regression, a BiLSTM with attention and a Transformer (all trained from scratch on HateXplain) against human rationales (IOU, AUPRC). Logistic regression matched or beat both neural models in 7 of 8 comparisons, significantly under SHAP ($p = 0.0006$ and 0.0020), traced to SHAP being exact for linear models but only sampled for neural ones.
- Measured ~41× lower training and 9.5× lower inference CO_2 emissions (CodeCarbon); BiLSTM explanations disagreed across random seeds (pairwise IOU 0.27–0.42) despite near-identical accuracy.

Raman Fuel Analysis System (abstract)

Hannah Nagpall (first author), M. R. Riazi, et al. · AIChE Annual Meeting 2026 · with Tec5USA R&D

- Lorentzian spectral model and NNLS composition estimation for gasoline, jet fuel, diesel and ethanol blends measured by Raman spectroscopy.
- Built the pure-component reference library by assigning Raman bands to molecular vibrational modes (e.g. ethanol C–C and C–O stretches, isooctane CH_3 modes) across the 300–1800 and 2700–3100 cm^{-1} regions, recording centre, width and height for 8–15 Lorentzian peaks per spectrum.
- Reported where each method fails: IHM was better only on jet fuel–ethanol (1.9% vs 7.2% mean error) and tied on diesel–ethanol, while isooctane–heptane was the hardest system (IHM 18.9%, NNLS 4.6%) because the two saturated alkanes have heavily overlapping bands.

Under Review

PrivAudit: An Interpretable Framework for Membership Inference Risk Auditing Using Composite Privacy Risk Scoring

Hannah Nagpall (sole author) · SN Computer Science · Zenodo · Code

- Designed an interpretable Composite Privacy Risk Score (five dataset-level signals) and a corrected Yeom-loss attack engine (balanced sampling, Youden's J threshold). Across 12 tabular benchmarks, CPRS tracked attack balanced accuracy (***r* = 0.846**, $p < 0.001$); ablation identified dataset size and feature-to-sample ratio as the dominant signals.
- L2 regularisation and Noise+PCA lowered CPRS by up to 38.6% (Heart Disease) with <0.01 change in attack balanced accuracy; released as an open-source Flask app.

Detecting Mixed-Authorship Academic Writing: A Leakage-Controlled Framework Combining Stylometric, Semantic, Readability, and Citation Features

Hannah Nagpall (sole author) · SN Computer Science · Zenodo · Code

- Three-class (Human / AI / Hybrid) detector on a 51-feature pipeline with family-aware cross-validation: held-out macro-F1 **0.611** (CV 0.530 ± 0.077; label-permuted baseline ~0.34). Citation-integrity features gave the largest ablation gain.
- Audited leakage explicitly: fitting preprocessing before the split inflated F1 by 0.08–0.12; a 5×5 cross-style study showed a 17.5-point generalisation gap.

RESEARCH EXPERIENCE

Research Collaborator — Behavioural Robustness of Clinical Prediction Models May 2026 – Present
Dept. of Computer Science, Texas A&M University–Kingsville

- Studying robustness and physiological consistency of temporal health prediction models under simulated clinical interventions (MIMIC-IV), quantifying prediction drift.
- Proposed the idea and independently designed a hybrid LSTM–Transformer model for temporal health data.

Research Assistant — Raman Spectroscopy Analysis Nov 2025 – Present
Dept. of Chemical & Natural Gas Engineering, Texas A&M University–Kingsville

- Ran the experiments in the Unit Operations Lab on a Tec5USA 785 nm Raman spectrometer (industry collaboration with Tec5USA): prepared 11 pure references, 8 binary systems and 2 ternary mixtures volumetrically (gasoline 87/89/93, diesel, jet fuel, ethanol, isooctane, heptane, benzene, toluene) and cross-checked with a density meter.
- Wrote my own analysis software: AsLS fluorescence-baseline removal, two-stage global Lorentzian fitting (soft-L1 then L2; R^2 ≥0.99 on pure components), peak assignment, and NNLS unmixing of mixture composition.
- NNLS recovered binary compositions with a **mean absolute error of 4.4 percentage points** (median 3.5; 18 mixtures, 6 systems) and beat an Indirect Hard Modeling baseline on 4 of 6 systems; fitted IHM parameters stayed near identity, indicating linear mixing. Abstract accepted at the AIChE Annual Meeting (Nov 2026).

Research Intern — Crystal Growth (Physics), NSF-INSPIRE Project Oct 2023 – Jul 2024
Alabama A&M University (hybrid)

- Physics research on the properties, synthesis and applications of high-tech crystals.

INDUSTRY & ENTREPRENEURIAL EXPERIENCE

Backend Development Intern — GSPANN Technologies, Inc. May – Aug 2025

- Built Java / Spring Boot / MongoDB REST APIs (authentication, product catalogue, orders) for an e-commerce platform.

AI in Retail Technology Extern — The Home Depot (remote) May – Jul 2025

- Benchmarked 3 AI risk frameworks for retail supply chains; summary adopted as cohort reference material.

Founder — EduQuest Advisors

- Founded a college placement advisory platform for student tennis players. Live site: hannahnagpall05.github.io/EduQuest.

VC Industry Research & Pitch Desk Extern — energy-sector market analysis and pitch preparation Jul – Aug 2024
NRF Foundation Student Program — New York, attended on the Ray Greenly Scholarship Jan 2025
CodePath Technical Interview Preparation 101 — 50+ algorithm problems solved in Python Feb – Apr 2025

SELECTED PROJECTS

MedBuddy: AI-Assisted Health Platform — React, Flask / Python, PostgreSQL, LLMs

- Full-stack platform centralising lab-result insights, medications, wellness tracking, appointments and caregiver support, with an LLM-backed layer that turns lab data into plain-language insights.

StudySense: ESP32 Intelligent Study Assistant — ESP32, IoT, Python, AI/ML, web dashboard

- Hardware–software system that runs Pomodoro sessions and senses the study environment, streaming data to the cloud for tracking.
- Web dashboard with study analytics and personalised recommendations, linking physical sensing to data-driven study habits.

StudyBuddy: Gamified Learning Platform — React, Flask / Python, PostgreSQL, LLMs

- Personalised study plans, interactive quizzes and flashcards, with progress-based rewards to sustain engagement.

TOEFL Quest: TOEFL iBT Practice Platform — HTML, CSS, JavaScript, Flask / Python

- Multi-mock TOEFL iBT preparation web app with section-wise practice for Reading, Listening, Speaking and Writing, real HTML5 audio, and speech-recognition-based speaking scoring. Live: toefl-quest.netlify.app.

TECHNICAL SKILLS

Machine Learning & Research:	Robustness / Drift Analysis, Explanation Faithfulness, Membership-Inference Auditing, Leakage-Controlled Evaluation, LLM Agents, Spectral Signal Processing
ML Frameworks & Libraries:	scikit-learn, XGBoost, SHAP, LIME, pandas, NumPy, SciPy, matplotlib, NNLS
Programming:	Python, Java, C++, SQL, HTML, CSS
Backend & Databases:	Spring Boot, REST, Flask, MongoDB, PostgreSQL, MySQL
Tools:	Git, GitHub, AWS, Ollama (local LLMs)
Human Languages:	English, Hindi

SCHOLARSHIPS

Texas A&M University–Kingsville

- Transfer Excellence Scholarship · C.J. Davidson Scholarship · John R. Bickham Engineering Scholarship · Honors College Scholarship · McCormick Leadership Scholarship · NCAA Division I Tennis (athletic) Scholarship

Alabama A&M University

- NCAA Division I Tennis (full athletic) Scholarship · Presidential Scholarship
- **External:** NRF Ray Greenly Scholarship, NRF Foundation Student Program, New York (Jan 2025)

HONOURS & ACTIVITIES

- **NCAA Division I Scholar-Athlete** (women's tennis), Alabama A&M, 2023–2025; **HBCU National Champion**. Project Q2 Global Research Hackathon, Participant (2026).
- Volunteer: tennis coach for people with disabilities (DreamCourt) · Samidha Welfare Foundation · old-age home. Certifications: Intro to ML (Kaggle) · Python (Google) · GCI World – AI & Data Science (University of Tokyo).